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UNIT - 1 (INTRODUCTION)

ONE SHOT VIDEO

SOFTWARE ENGINEERING

⊙ Topics :-

→ Software Engineering : Characteristic [2021-22]
→ Software Crisis.
→ Software Engineering Vs Traditional Engineering Process
→ S.D.L.C [2021-22/23 AKTU]
→ Water Fall Model [2022-23 AKTU]
→ Prototype Model [2021-22 AKTU]
→ Spiral Model [2021-22 AKTU]
→ Evolutionary Development Models
→ Iterative Enhancement Models

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⊙ Software Engineering :-

→ Software Engineering is the branch of computer science that deals with the design, development, testing and maintenance of software applications.

→ Software Engineers apply engineering principles and knowledge of programming language to build software solutions for end user

→ Software engineering is a discipline whose aim is the production of fault free software that satisfies the user's needs and that is delivered on time & within budget

→ Objective of Software Engineering :-

- To understand user conceptual models and development of better specification.
- To improve design language and reusable code
- To satisfy the user's requirements.
- To achieve low maintenance and production cost.
- To provide the software within buget and time
- To achieve high performance.

→ Principles of Software Engineering :-

1. Modularity : Breaking the software into smaller, reusable components that can be developed and tested independently

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2. Abstraction :- Hiding the implementation detail of a component and exposing only the necessary functionality to other part of the software
3. Encapsulation :- Wrapping up the data and function of an object into an single unit, and protecting the internal state of an object from external modification
4. Reusability :- Creating component that can be used in multiple projects, which can save time and resources
5. Maintenance : Regularly updating and improving the software to fix bugs, add new features and address security vulnerabilities.
6. Testing : Verifying that the software meets its requirements and is free of bugs.
7. Design Patterns : Solving recurring problems in software design by providing templates for solving them
8. Continuous Integration & Deployment : Continously integrating the code changes and deploying them into the production environment.

Software Characteristic :-

→ Software :- is a defined as a collection of computer program, procedures, rules and data.

→ Software characteristic is classified into six major components :-

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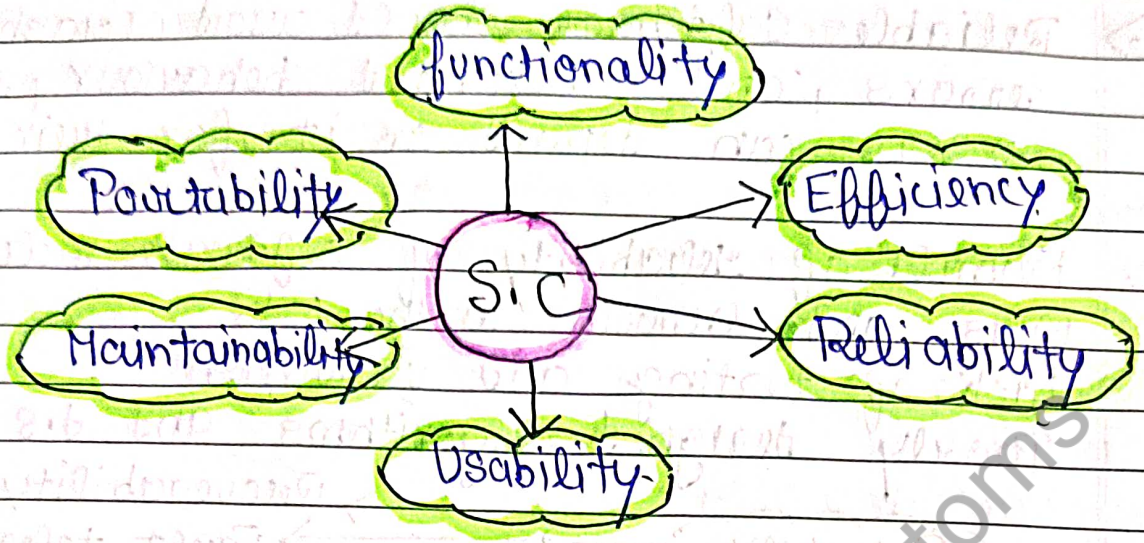
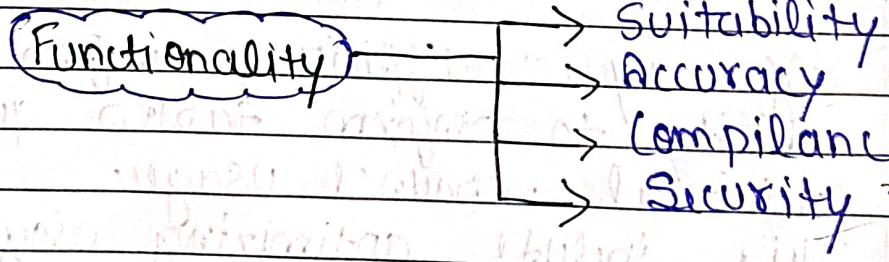


1. Functionality :- This refers to the ability of the software to perform its intended tasks or functions.
→ It includes features such as calculations, data processing, user interface interactions and more
→ Good software functionality meets user requirements effectively and efficiently.

Examples of functionality in software include :-
- Data storage and retrieval
- Data processing and manipulation
- User interface and navigation and many more



2. Reliability :- Reliability relates to the ability of the software to perform consistently and predictably under various condition

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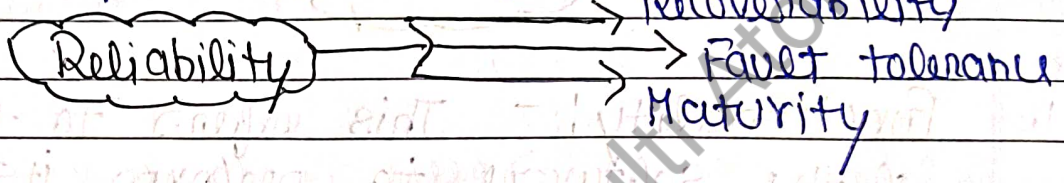
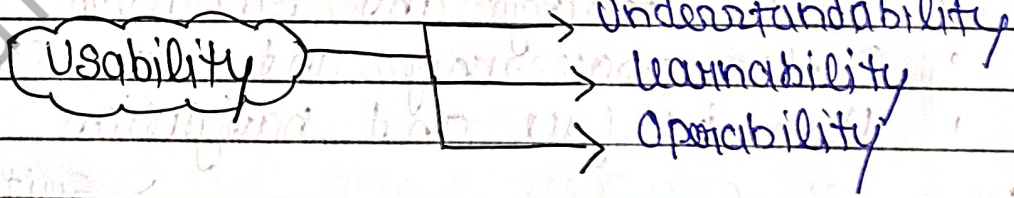
→ Reliable software should avoid crashes, errors, and unexpected behaviour, ensuring users can depend on it for their tasks.

Examples of reliability of software include:-
- Bugs and errors in the code
- Lack of testing and validation
- Poorly designed algorithms and d.s.



3. Usability :- Usability focused on how easy and intuitive it is for users to interact with the software.
→ This includes aspects like user interface design, navigation, accessibility, and overall user experience.
→ The amount of effort or time required to learn how to use the software



4. Efficiency :- Efficiency refers to the software ability to perform tasks quickly and with minimal resource usage.
→ This include optimizing algorithms, utilizing system resources effectively, and reducing unnecessary overhead.

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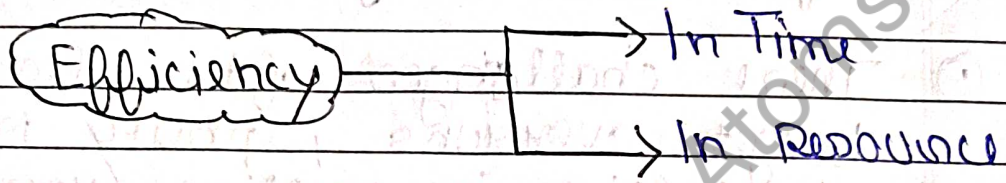
Example of Factors that can effect the efficiency of the software include:

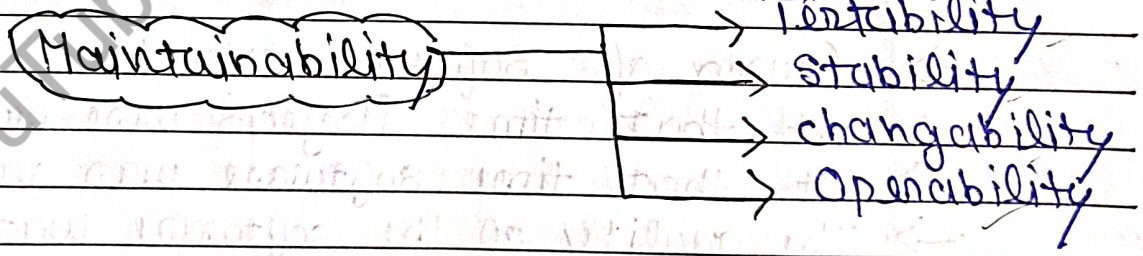
- Poorly designed algorithms and data structures
- Unnecessary processing and computation
- Unoptimized code.



5. Maintainability :- Maintainability measures how easy it is to maintain and update the software over its lifecycle.

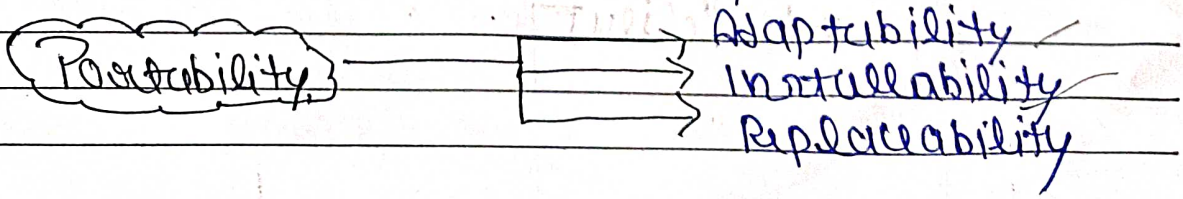
→ It includes factors like code readability, modularity, documentation and the use of best practices.

→ Maintainable software is easier to debug, enhance and scale.



6. Portability :- Portability refers to the software ability to run on different platforms or environments without major modifications.

→ This includes consideration for operating system hardware architecture, and dependencies.



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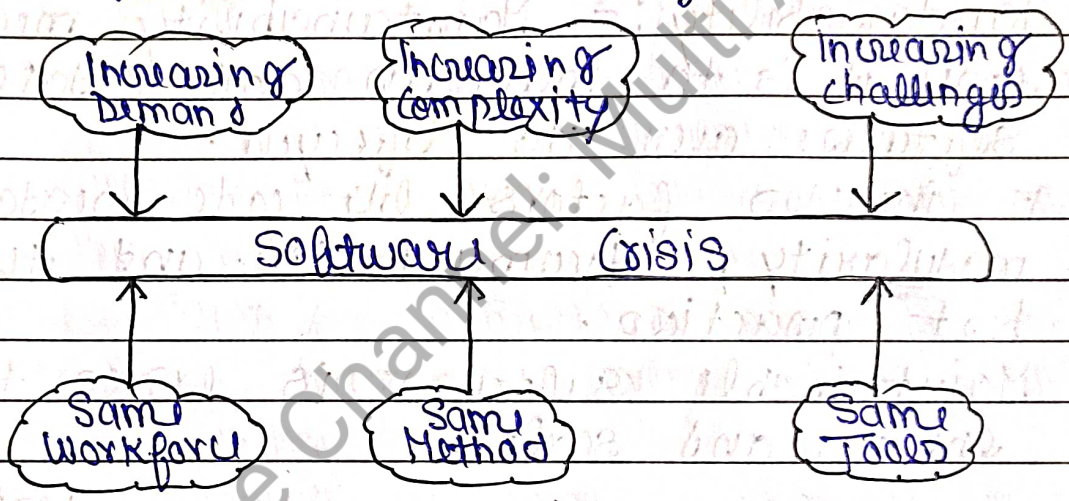
⇒ Software crisis :-

- The term "software crisis" refers to a set of challenges and issues that arise in the development and maintenance of software system.
- These challenges often lead to delays, budget overruns, quality issues, and difficulties in meeting user requirements.



⇒ Causes of Software crisis :-

→ At that time Projects were running overtime
→ At that time software was very inefficient
→ The quality of the software was low quality
→ Software often did not meet user requirement
→ At that time software was never delivered
→ Non-optimal resource utilization.
→ The software complexity is harder to change.
→ Lack of communication between Developer and client

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Solution of Software Crisis:

→ Reduction in software over budget
→ The quality of the software must be high.
→ Less time is needed for a software project.
→ Experienced and skilled people working on the software project.
→ Software must be delivered.

⇒ Software Engineering vs Conventional Engineering

⊙ Software Engineering Process:-

⊙ Software Engineering process is a process that majorly involved computer Science, information technology and decrete mathematics.
⊙ It is mainly related to computers, programming and writing codes for building application
⊙ In software Engineering Process construction and development cost is low.
⊙ It can involve the application of new and untested elements in software projects.
⊙ In S.E.P most development effort goes into building new design and features.
⊙ It majorly emphasizes quality
⊙ Testing is an integral part of the development process.
⊙ Prototyping is common and helps to refine requirements
⊙ Maintenance and updates are necessary to keep software relevant.
⊙ It involve complex logic and algorithm

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- It is less expensive.
- Agile methodologies are commonly used in S.D.

- Conventional Engineering Process :-
- The conventional Engineering Process is a process that majorly involves science, mathematics, and empirical knowledge.
- It is about building cars, machine, hardware buildings, etc.
- It's construction and development cost is high
- It usually applies only known and tested principles to meet product requirements
- In conventional Engineering process, most development efforts are required to change old design
- It majorly emphanized mass production.
- Testing occurs mainly after product completion
- Prototyping is less common due to cost and time.
- Maintenance is typically scheduled or reactive
- It may have more complex physical properties to deal with
- It is more expensive.
- Ex Project management approaches.

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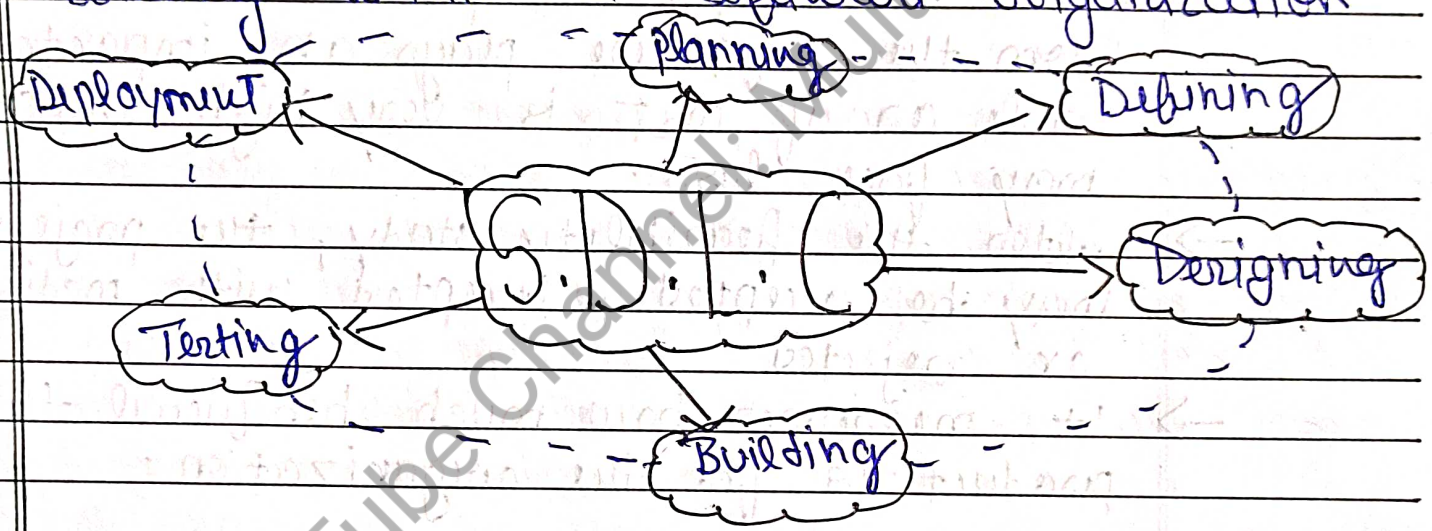
S.D.L.C

→ It stands for Software Development life cycle.
→ A software Development life cycle model is a pictorial and diagrammatic representation of the Software life cycle.
→ SDLC is used by the software industry to design, develop and test high quality softwares
→ SDLC is a process followed for software building within a software organization



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• Phase 1: Requirement and Planning Analysis:-
→ It is the first phase of SDLC in which all the necessary information is collected from the customer to develop the software as per their expectation
→ Some important question like: what is the need of software, who will be the end-user, what is the future scope of that software are discussed.
→ The main aim of this phase is to collect all the details of each requirement of the customer

so that the developer will clearly understand what they are developing and how to fulfill the customer's requirement

→ This phase gives a clear picture of what we are going to build.

• Phase 2 : Definning and Fearibility study :

→ It is the second phase of SDLC in which an organization discusses about the cost and benefits of the software

→ It is an important phase because profit from the software plays an important role as if cost is very high then company may face loss.

→ After the fearibility study, the project may be accepted, accepted with modification or rejected

→ It measures how much beneficial the product is for the organization.

• Phase 3 : Design :-

→ It is the third phase in which architects start working on logical designing of the software.

→ In this phase a SRS (system Requirement Specification) document is created which contains all logical details like how the software will look like, which language will be used, database design, modular design etc.

→ This phase provides a prototype of the final Product.

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• Phase 4 : Coding And Building :-

→ When the designing of the software is completed, then a group of developer start coding of the design using a programming language.

→ The interface of the software and all its internal working according to design phase is implemented in coding phase.

→ A number of developer code the modules and then all modules are arranged together to work efficiently.

→ It is the longest phase of SDLC

• Phase 5 : Testing :-

→ Once the software development is completed, then it is sent to the testers. The testing team start testing the functionality of the entire system.

→ In this phase, the software is checked for bugs or errors.

→ Whenever a bug is formed or found, then the software is resent to the coder to fix it and then overall software is re-tested.

→ This is done to verify that the entire application works according to the customer requirement.

• Phase 6 : Deployment and Maintenance :-

→ After overall testing of the software and after checking that is bug free, then the software is launched and available for the user to use it.

→ Even after deployement of the software, if any

bugs or errors are still found then the software is re-evaluated by the maintenance team and then it is re-deployed with a new version

Maintenance :-

⇒ The Maintenance team look over the software usage and user's feedbacks

⇒ Maintenance is necessary to eliminate errors in the system during its working life and to tune the software.

⇒ The bug fixing, upgrade and enhancement of the software is looked over by the maintenance team.

⇒ Need of S.D.L.C :-

- Increased visibility of the development process for all stakeholders involved.
- Efficient estimation, planning and scheduling.
- Improved risk management and cost estimation
- Systematic software delivery and better customer satisfaction

⇒ Models :

(1) Water Fall model
(2) Prototype model
(3) Spiral model
(4) Evolutionary Development Models
(5) Iterative Enhancement Models :

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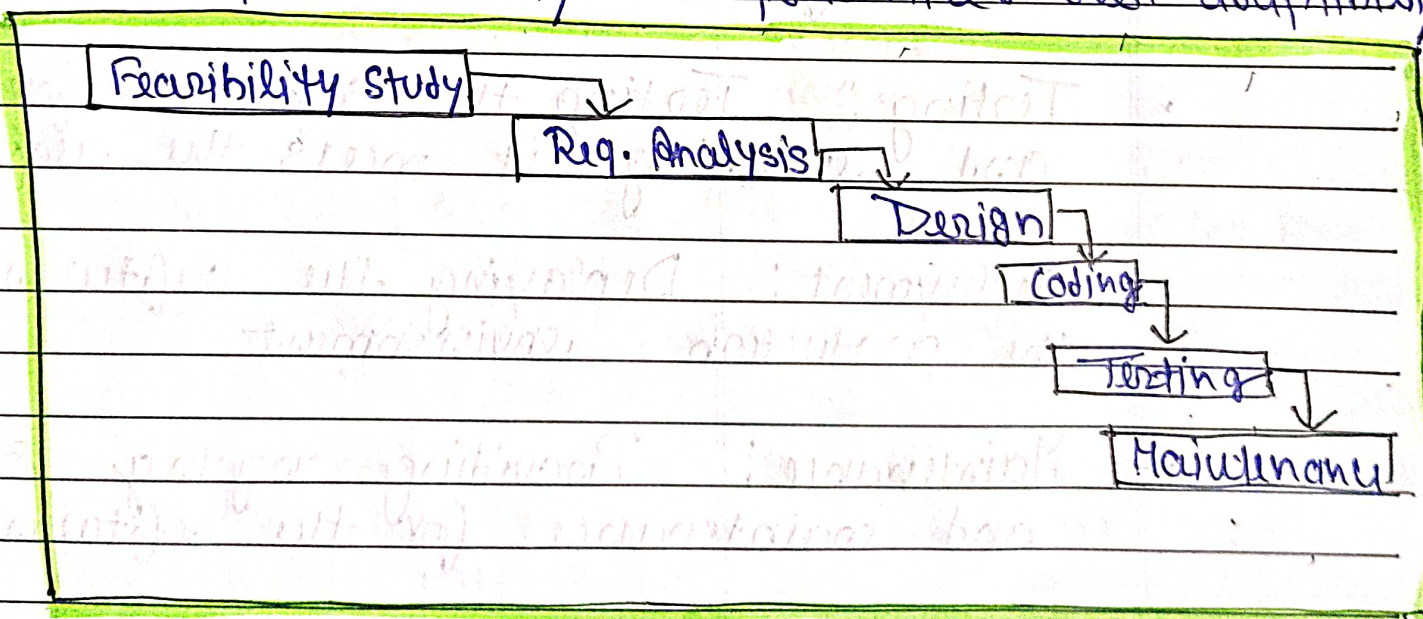
⇒ WATERFALL MODEL

-> Developed in the 1970s by Winston W. Royce.
-> Inspired by manufacturing and construction processes, where each step relies on the completion of the previous one.
-> It is a simplest SDLC in which phases are organised in a linear and sequential order.
-> It is called waterfall model as its diagrammatic representation resembles as like a waterfall, also known as classical life cycle model.
-> This type of model is basically used for the small to medium - sized project with clear, well - defined requirements.
-> When the technology and tools to be used are well - known and stable.
-> Projects where minimal changes are expected during the development process, and predictability is prioritized over adaptibility.



Waterfall Model

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⇒ Phases :-

The waterfall model consist of distinct phases that are executed sequentially, with each phase building upon the previous one.

- Feasability study :- In this phase of water fall model in which an organization discusses about the cost and benefits of the software.
- Requirement Analysis :- Gathering and documenting software requirements
- Design :- Creating a detailed design based on the requirements.
- Implementation : Coding and developing the software based on the design
- Testing : Testing the software for defects and ensuring it meets the requirement
- Deployment : Deploying the software to the production environment
- Maintenance : Providing ongoing support and maintenance for the software

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Adv :-
- Simple and easy to understand to use.
- Easy to manage due to the rigidity of the model.
- Phases are passed and completed one at a time.
- Works well for smaller projects where requirements are very well understood.
- Clearly defined stages
- Easy to arrange tasks.

Disadv :-
- No working software is produced until late during the life cycle.
- Hight amount of risk and uncertainty
- Not a good model for complex and O.O project
- It is difficult to measure progress within stages
- Cannot accomodate changing requirements

⇒ PROTOTYPE MODEL

⇒ Most of the customers are not sure about the functionality they require from the software, as a result the final S/W is not according to exact demand.

⇒ It is an iterative approach, which involves developing an early working model of the software based on the currently known requirements with limited functionalities, low reliability and untested performance.

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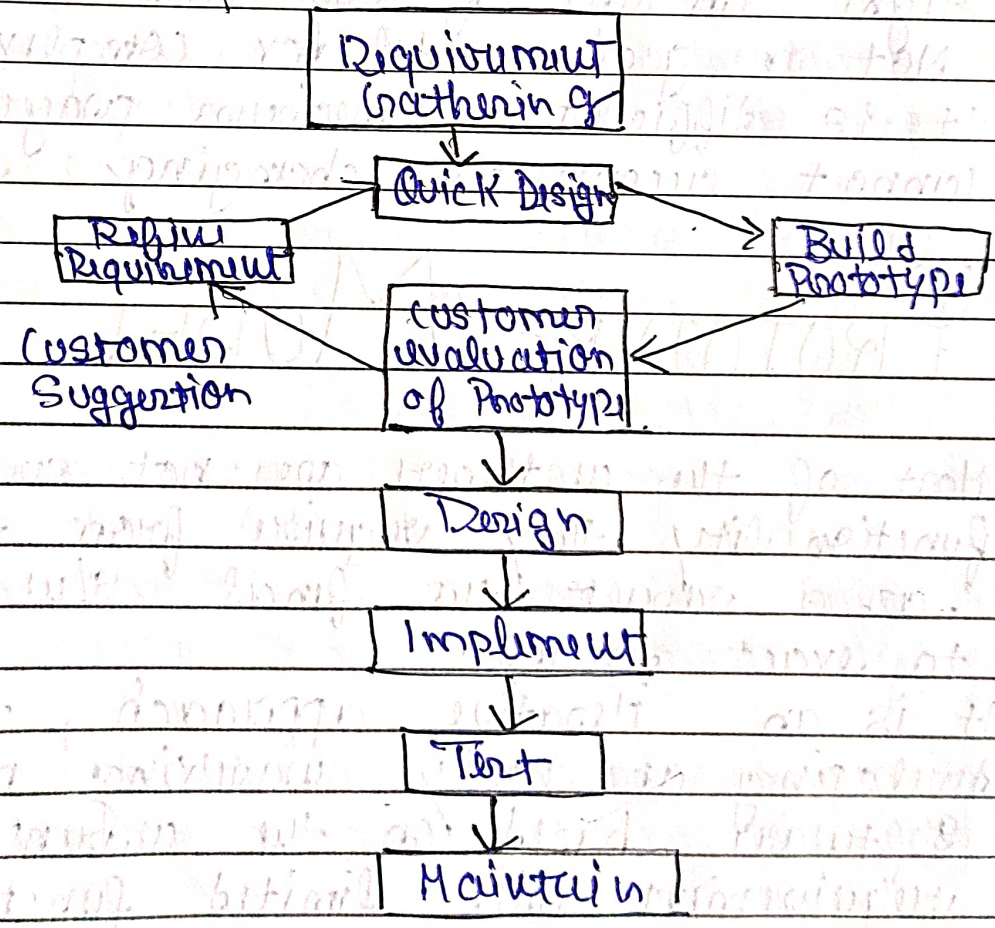
→ Refining it through user feedback and repeating the process until a satisfactory solution is achieved.

→ Then shown to the user, as per the feedback of the user prototype is rebuilt and modified and again show to the user, the process continue till the customer is not satisfied.

→ After this process the final SRS document is developed.

→ Developing the prototype help in building the actual design.



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1. Requirement Gathering : This is the initial step in designing a prototyping model.
→ In this phase, user are asked about what they expect or what they want from the system

2. Quick Design :- This model cover the basic design of the requirement through which a quick overview can be easily described

3. Build a Prototype :- This step helps in building an actual prototype from the knowledge gained from prototype design.

4. Initial User Evaluation :- This step describes the preliminary testing where the investigation of the performance model occurs, as the customer will tell the strength and weakness of the design, which was sent to the developer.

5. Refining Prototype : If any feedback is given by the user, then improving the client's response to feedback and suggestions, the final system is approved.

6. Design :- After the approval, creating a detailed design based on the refining prototype

7. Implement :- Coding and developing the software based on the design

8. Test: Testing the software for defect and ensuring it meets the requirement

9. Maintain:- Providing ongoing support and maintainence for the software.

⇒ Types Of Prototype:-

⇒ There are four types of Prototyping Models, which are described below.

1. Rapid Throwaway Prototyping
- This technique offers a useful method of exploring ideas and getting customer feedback for each of them
- In this method, a developed prototype need not necessarily be a part of the accepted prototype.
- Customer feedback helps prevent unnecessary design faults and hence, the final prototype developed is of better quality.

2. Evolutionary Prototyping
- In this method the prototype developed initially is incrementally refined based on computer feedback till it finally gets accepted.
- In comparison to Rapid Throwaway prototyping, it offers a better approach that save time as well as effort.

- This is because developing a prototype from scratch for every iteration of the process can sometimes be very frustrating for the developers

3. Incremental Prototyping :-
- In this type of incremental prototyping, the final expected product is broken into different small pieces of prototypes and developed individually.
- In the end, when all individual pieces are properly developed, then the different prototypes are collectively merged into a single final product in their predefined order.

4. Extreme Prototyping :- This method is mainly used for web development. It consist of three sequential independent phases :
- In this phase, a basic prototype with all the existing static pages is presented in HTML format.
- In the 2nd phase, Functional screens are made with a simulated data process using a prototype services layer.
- This is the final step where all the services are implemented and associated with the final prototype.

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⇒ Adv :
→ Reduce the risk of incorrect user requirement
→ Good where requirement are changing / uncommitted
→ Regular visible process aids management
→ Support early product marketing
→ Reduce maintenance cost
→ Error can be detected much earlier as the system is made side by side

⇒ Disadv :
→ Required extensive customer collaboration
→ Difficult to know how long the project will last.
→ Easy to fall back into the code and fix without proper requirement analysis, design customer evaluation and feedback
→ It tools are expensive
→ It is a time consuming process

SPIRAL MODEL :

→ Barry boehm recognised the project risk factor into a life cycle model, the result was the spiral model, which was presented in 1986.

→ Spiral model is a combination of a waterfall model and iterative model.
→ Thats why it is also known as meta model (model about model)

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→ Each phase in spiral model begins with a design goal and ends with the client reviewing the progress.

→ The development team in spiral - SDLC model starts with a small set of requirement and goes through each development phase for those set of requirements.

→ The software engineering team adds functionality for the traditional required in every-increasing spirals until the application is ready for the production phase.

→ A spiral model is divided into a number of framework activities, also called task regions.

→ The spiral model in S.E consists of four primary phases which are repeated in iterations throughout the development process.

1. Planning: In this phase, project objectives, requirements, constraints, and risks are identified.

→ A preliminary project plan is created based on these inputs, outlining the resources, timelines and deliverables for the project.

→ In this phase the scope of the project is determined and a plan is created for the next iteration of the spiral.

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2. Risk Analysis :- This phase involves a thorough analysis of identified risks from the planning phase.

→ In the risk analysis phase, the risk associated with the project are identified and evaluated

→ Risk are categorized based on their impact and likelihood of occurunce

3 Engineering (Development and Testing): In this phase, the software is developed based on the requirement gathering in the previous iteration.

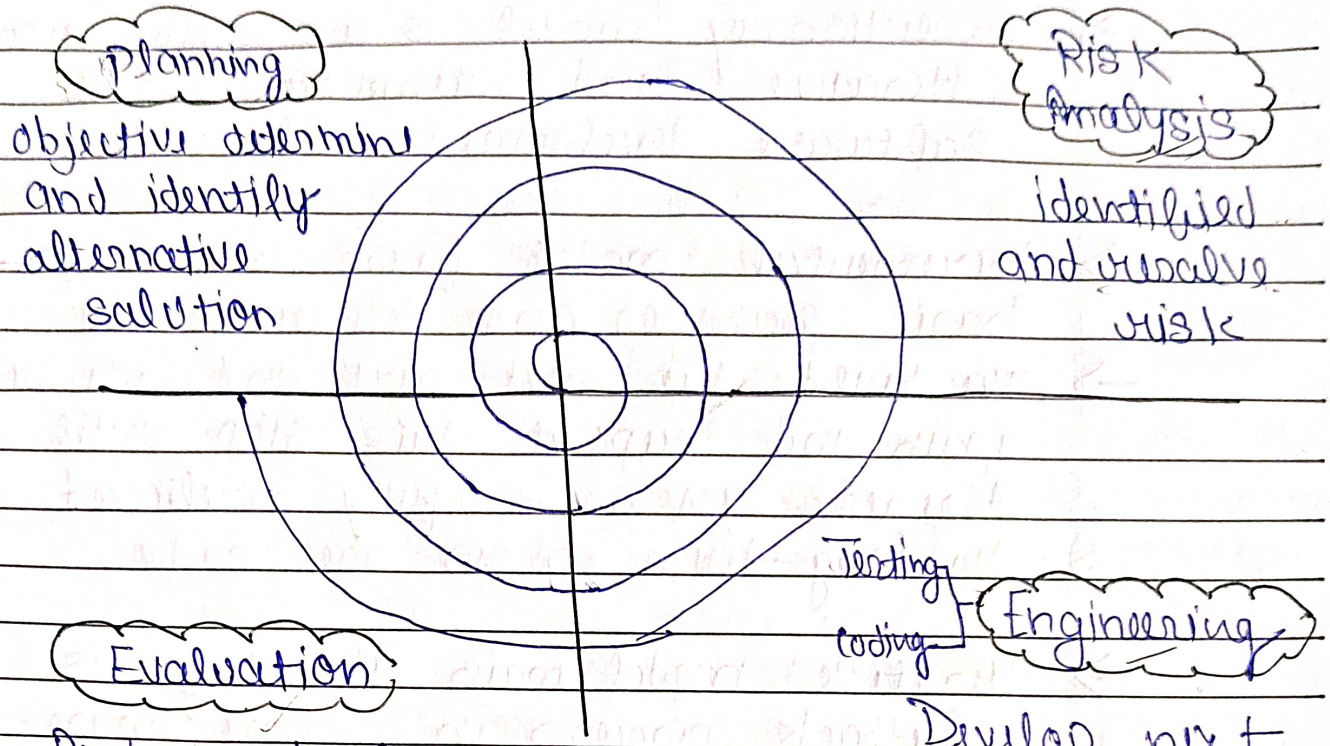
→ This includes designing, coding, integrating components, and testing the software.

4 Evaluation (Review and Feedback): In this phase, the software is evaluated to determine if it meets the customer requirements and if it is of high quality.

→ After completing each iteration of development, the software is evaluated through reviews, testing, and feedback from stackholders

→ Stackholder provide feedback on the software strengths, weakness and areas for improvement, which is used to make inhancement and adjustment in subsequent iterations

There four phases are iteratively repeated throughout the spiral model.



Spiral Model

Adv :-
→ Risk handling
→ Good for large project
→ Flexibility in requirement
→ Customer Satisfaction
→ Iterative and incremental Approach

Disadv :-
→ Complex → Difficulty in time management
→ Expensive. → Time consuming. → Resource Intensive
→ Too much dependability on Risk analysis.

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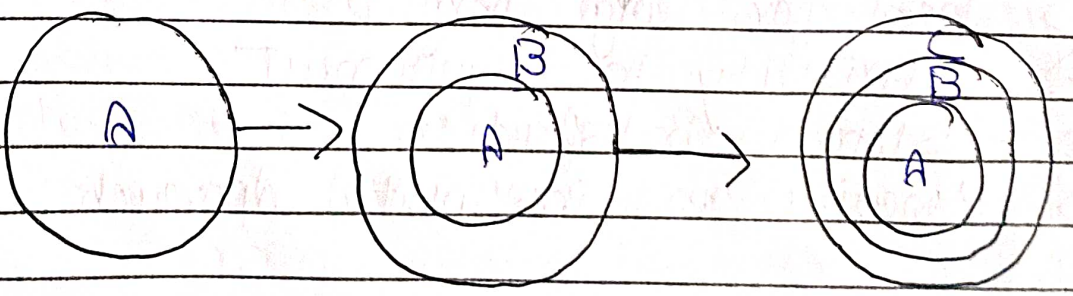
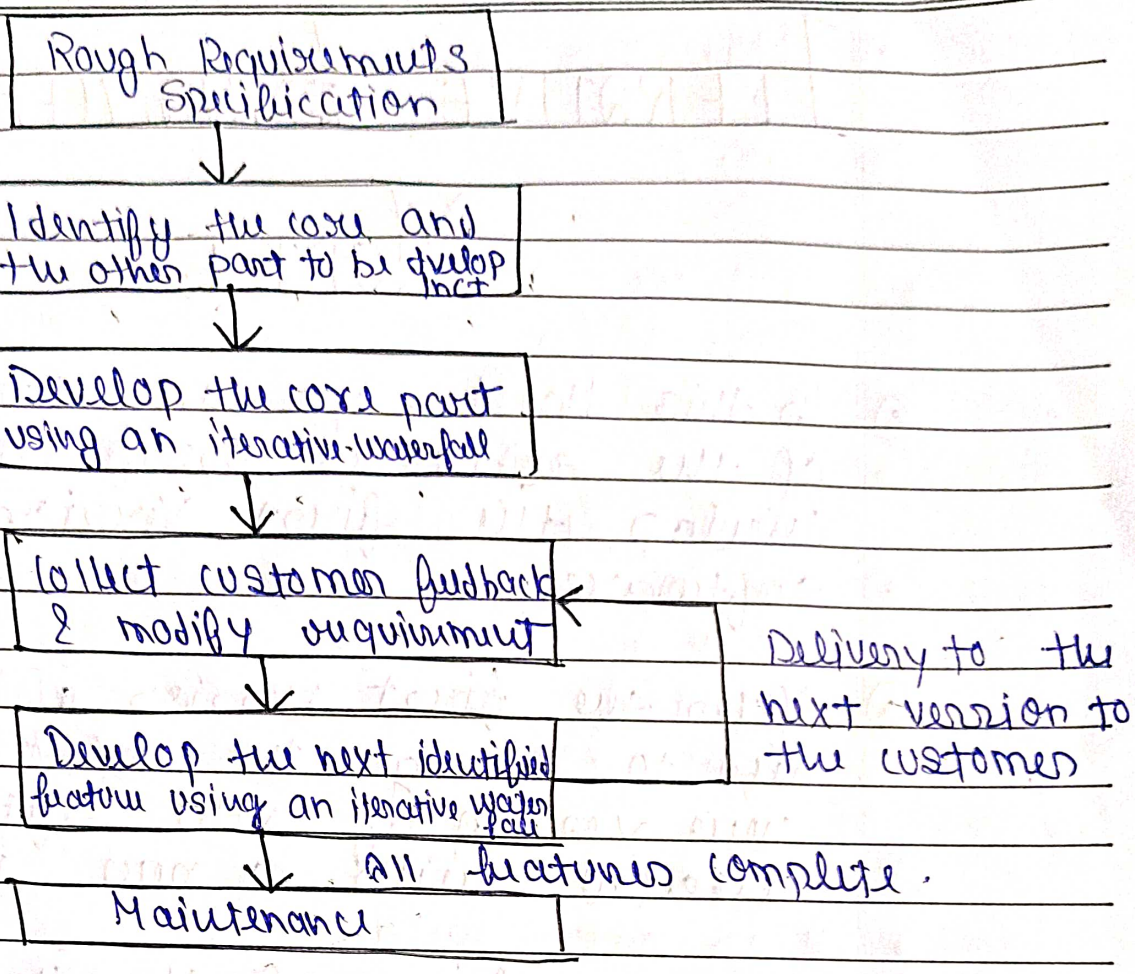
EVOLUTIONARY DEVELOPMENT MODEL

=> Evolutionary model is a combination of iterative and incremental model of software development life cycle.

=> Incremental modal first implement a few basic features and deliver to the customer.
=> The build the next part and deliver it again and repeat this step until the desired system is fully reliazed.
=> No long-term plans are made

=> Iterative model main advantage is its feedback process in every phase.
=> Also known as "Design a little, build a little, test a little, deploy a little model.
=> This model is also known as the successive versions model.





Adv :

→ Customer requirements are clearly specified
→ Risk analysis is better
→ It support changing environment
→ Initial operating time is less.
→ Better suited for large mission - critical projects

Disadv :

→ Not suitable for smaller projects
→ Cost
→ Highly skilled resources are required

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ITERATIVE ENHANCEMENT MODEL

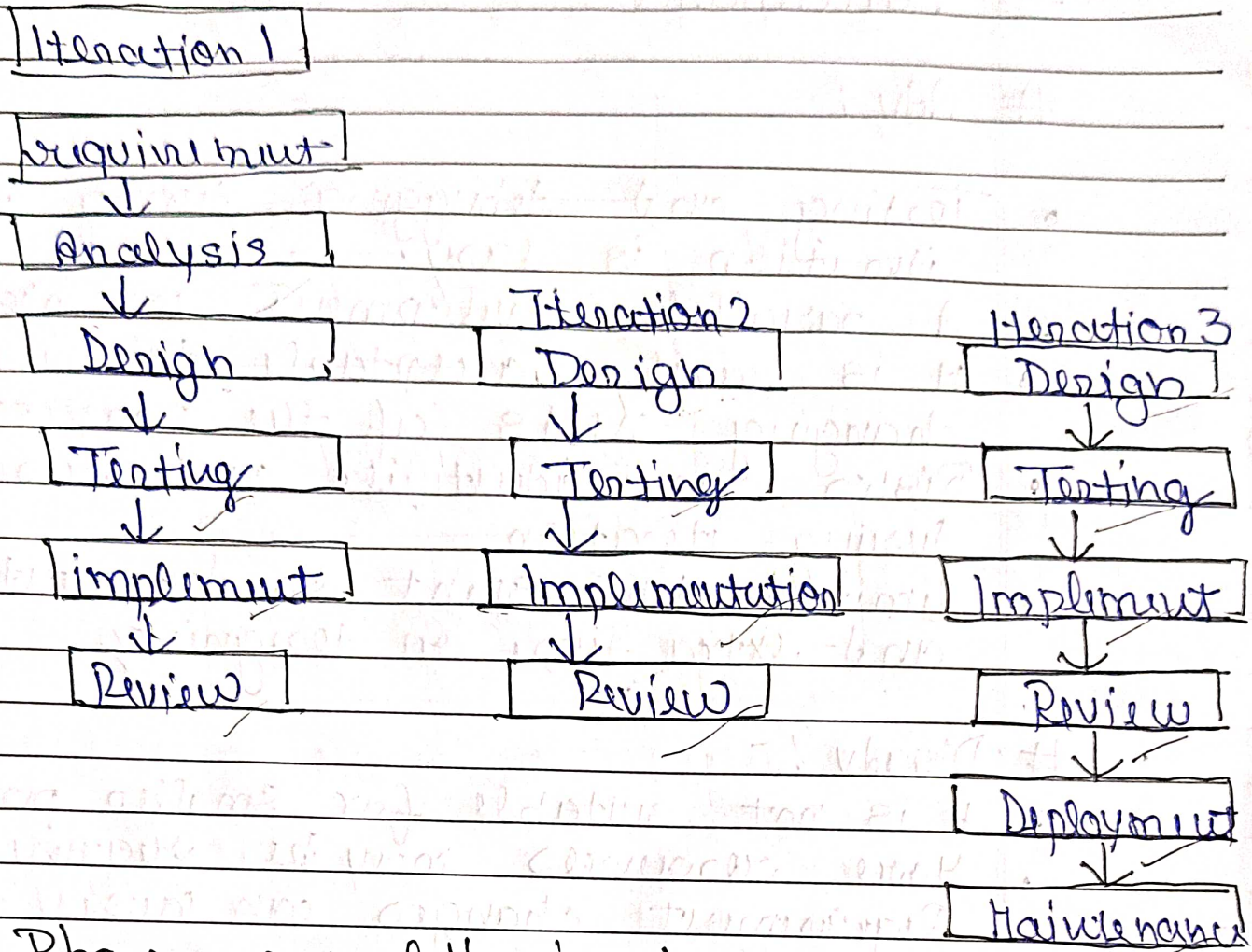
① In this Model, you can start with some of the software specification and develop the first version of the software.

② After the first version if there is a need to change the software, then a new version of the software is created with a new iteration.

③ Every release of the iterative model finishes in an exact and fixed period that is called iteration

④ The iterative model allows the accessing earlier phases, in which the variations made respectively.

⑤ The final output of the project renewed at the end of the SDLC process



⇒ Phases are following :-
⇒ Requirement gathering & analysis
⇒ Design
⇒ Implementation
⇒ Testing
⇒ Deployment
⇒ Review :- In this phase, after the product deployment, review phase is performed to check the behaviour and validity of the development product and if there are any error found then the process starts again from the requirement gathering.

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→ Maintenance

Adv :

- Testing and debugging during smaller iteration is easy.
- A parallel development can plan.
- It is easily acceptable to ever-changing needs of the project.
- Risks are identified and resolved during iteration
- limited time spent on documentation and extra time on designing

Disadv :-

- It is not suitable for smaller projects
- More resources may be required.
- Requirement changes can cause over budget.
- Project completion date not confirmed because of changing requirement

⇒ Software Quality Attributes :-

☆ Software Quality Attributes are features that facilitate the measurement of performance of a software product by software Testing professional.

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Software quality attributed are :-

1] Correctness : A system is functionally correct if it behave according to its functional requirement specification

2] Reliability : A system is said to be reliable if it gives desired output even in case of change of component (like hardware) or in case of overload conditions

3] Robustness : It explain how does a system behave in situations not specified by its requirement i.e incorrect input, hardware failure, loss of power

4] Interoperability :- It describes the ability of two system to engage in the exchange of information via certain interfaces

5] Performance :- This attribute pertains to the ability of a software-driven system to conform to timing requirement

6] Efficiency :- It is the ability of software system to fulfill its purpose with best possible utilization of necessary

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resources such as time, storage, transmission channel, peripherals etc

⑧ Security :- The software's ability to protect against unauthorized access, data breaches, and other potential threat.

⑨ Portability :- The ability of the softw are to operate on different platforms or environments without significant modification

⑩ Modularity : The degree to which the software's component to be organised into separate, manageable units that can be independently developed or updated

⑪ Testability :- The ease with which the software can be tested to ensure it meets its requirement and performs as expected

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